Rajeev Sharma



Software Engineer | Machine Learning | Full-Stack Development

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rajeev-dev.vercel.app: portfolio, projects, bachelor thesis and references

Profile

Computational Engineering Science graduate (B.Sc., TU Berlin) with nearly two years of R&D experience as a working student, spanning Java, Python, Linux-based embedded systems, hardware/software interfaces (UART, SPI), medical sensor data, signal processing and machine learning. Independently designed and implemented an end-to-end bachelor thesis pipeline (grade 1.3) from real-world biosignal acquisition to CNN training, quantisation and BrainChip Akida neuromorphic simulation. Currently strengthening computer-science foundations (Harvard CS50x) and backend/full-stack skills through personal projects.

Education

B.Sc. Computational Engineering Science

Technische Universität Berlin | 2017–09/2025 | Final grade: 2.8

Specialisation: Informationstechnik im Maschinenwesen (information technology in mechanical engineering). Selected grades: bachelor thesis 1.3; Engineering Data Analytics with R 1.7; Fundamentals of Automation Engineering 1.7; System Programming in C 2.0.

Studienkolleg, T-Kurs | Technische Universität Berlin | 2015–06/2016

Feststellungsprüfung (university entrance qualification for technical degree programmes), grade 2.0.

Professional Experience

Working Student – Research & Development (part-time, alongside studies)

SectorCon Ingenieurgesellschaft mbH, Berlin | 11/2023–10/2025

- Co-developed a Java component that reconstructs byte-level SPI data and converts it into structured JSON.
- Researched UART/SPI communication and Java libraries (jSerialComm, Pi4J) on Linux and Raspberry Pi.
- Re-implemented the company's internal publish/subscribe functionality in Java.
- Developed Java functions that retrieve and evaluate modem and Wi-Fi status via Linux system commands.
- Wrote and executed JUnit tests; set up remote JVM debugging and remote execution on Linux.
- Worked with Git/GitLab, terminal workflows and technical documentation in the engineering team.

Bachelor Thesis: Signal Quality Analysis of Medical Sensor Signals Using Neural Networks and Implementation on a Neuromorphic Chip

SectorCon Ingenieurgesellschaft mbH, Berlin, in cooperation with TU Berlin | 03/2025–07/2025 | Grade: 1.3

- Independently collected pulse-wave data with a piezoelectric sensor and LabVIEW (250 Hz, about 18,000 labelled samples); preprocessing, feature extraction and time-frequency analysis (FIR filtering, CWT, STFT) in Python and MATLAB.
- Developed a CNN with TensorFlow/Keras to classify signal quality with **81.49% test accuracy** (AUC 0.89); handled class imbalance with oversampling and evaluated with precision, recall, F1, ROC/AUC and confusion matrices.
- Quantised (QuantizeML 8-4-4), fine-tuned and converted the model for the neuromorphic Akida platform: **79.53% test accuracy** in the Akida runtime on a CPU host, less than 2 percentage points below the original model.

Skills

Programming: Python, Java, C, SQL, JavaScript, TypeScript, R, Bash, MATLAB

AI and Machine Learning: TensorFlow/Keras, scikit-learn, imbalanced-learn, CNNs, model evaluation, quantisation, Akida MetaTF/cnn2snn (simulation)

Data and Signal Processing: pandas, NumPy, SciPy, PyWavelets, FIR filtering, CWT/STFT, feature extraction, data cleaning and integration, KPI analysis

Backend and Databases: REST APIs with FastAPI, Flask and Spring Boot (personal project), JSON, publish/subscribe, SQL, PostgreSQL, SQLite

Frontend and Visualisation: React, Vite, Tailwind CSS, Three.js, R Shiny, Plotly, Matplotlib

Embedded and Communication: Linux, Raspberry Pi, SPI/UART, jSerialComm, Pi4J, LabVIEW

Tools: Git/GitHub/GitLab, feature branches, JUnit, remote JVM debugging, CI/CD with GitHub Actions, Vercel; basic: pytest, Docker, AWS EC2, Azure

Exposure: ROS/ROS 2, MQTT, Kafka, OPC UA (research level)

Personal Projects

Sensor-to-Cloud Signal Platform (in development)

Java, Spring Boot, PostgreSQL, JUnit, Testcontainers, GitHub Actions | since 10/2026

- Extending my bachelor thesis into an end-to-end sensor-data platform using public PPG data and a telemetry simulator.
- Spring Boot REST API with PostgreSQL persistence (JPA/Hibernate), JUnit and Testcontainers tests, and a CI pipeline in GitHub Actions.
- Planned: signal analysis and model serving in Python, MQTT telemetry simulation and a results dashboard.

AMR Cleaning Digital Twin (in progress)

React, TypeScript, Vite, Three.js

- Browser-based 3D visualisation of my cleaning-robot concept from university, developed with feature branches, ESLint and Prettier.

wohnmatch-AI (in progress)

Python, FastAPI, Docker

- Early-stage housing-search backend with a containerised FastAPI service and OpenAPI docs; matching and frontend features are planned.

Personal Portfolio Website (completed)

React, JavaScript, Vite, Tailwind CSS, GitHub Actions, Vercel | rajeev-dev.vercel.app

- Designed and published a responsive portfolio with reusable React components and a contact form; lint and build checks in CI, automatic deployment on Vercel.

Academic Projects

Automotive Failure Analysis | R and Shiny | Team of 4 | Grade: 1.7

- Cleaned and integrated production, bill-of-materials, registration and geodata, analysed failure KPIs and co-developed an interactive Shiny application for the evaluation.

Autonomous Mobile Cleaning Robot | Conceptual design, individual work | Grade: 2.0

- Defined 44 requirements and designed a modular system architecture, a route-planning concept for a 534 m² hall and CAD layouts.

Further Education

CS50x: Introduction to Computer Science | Harvard University (online)

In progress since 10/2026, expected completion 01/2027.

Other Experience

Retail Assistant, Primark, Berlin | 05/2026–present

Part-time retail role (30 h/week) while applying for software and ML positions.

Career break | 12/2025–04/2026: extended stay in Nepal.

Languages

German: C1 | English: C1 (fluent) | Nepali: native | Hindi: fluent | French: A1